LESSON 1.4

Math is Explaining and Sharing

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can construct an argument using equations, drawings, or words, and use volume to defend a recommendation.

WHY THIS MATTERS Yuzuki is buying her favorite breakfast cereal. She can buy three large-sized boxes or two jumbo-sized boxes for the same price. Which option should she pick — and can you convince a classmate you are right? Each large-sized box holds 240 cubic inches of cereal and each jumbo-sized box holds 400 cubic inches.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
argument *Spanish: argumento*	A chain of reasons — equations, drawings, or words — used to defend your thinking.	"Two jumbo boxes hold more, because 400 × 2 = 800 > 720."
conjecture *Spanish: conjetura*	A mathematical statement you believe is true but have not yet proven.	"A bigger box ALWAYS holds more cereal" — believed, not yet proven.
counterexample *Spanish: contraejemplo*	A single example that shows a statement is not always true.	One jumbo box sold half full of air breaks "bigger always holds more".
volume *Spanish: volumen*	The amount of space a solid figure takes up, measured in cubic units.	8 in × 3 in × 10 in = 240 cubic inches of cereal space.
cubic unit *Spanish: unidad cúbica*	A cube with edges 1 unit long — the unit used to measure volume.	One cube, 1 unit on every edge — the block volume is counted in.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Suppose each large box is 8 inches long, 3 inches wide, and 10 inches tall. Its volume is 8 × 3 × 10 = 240 cubic inches, so three large boxes hold 240 × 3 = 720 cubic inches.

1. Suppose each jumbo box is 10 inches long, 4 inches wide, and 10 inches tall. Its volume is 10 × 4 × 10 = 400 cubic inches, so two jumbo boxes hold 400 × 2 = 800 cubic inches.
2. I recommend buying two jumbo boxes because 800 cubic inches is greater than 720 cubic inches — the total volume is greater, so there is more cereal for the same price.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Not every good reason is an equation. Finish this argument: “I recommend buying two jumbo boxes because …”

1. _____
2. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- An argument using _____ is stronger here because _____.
- I recommend _____ because _____ .
- A reason that is not about volume is _____ .
- Explaining helps because _____ .

Word bank argument | conjecture | counterexample | volume | cubic unit

WATCH OUT!

A common mistake

Reporting volume without cubic units

LESSON 1.4

Math is Explaining and Sharing

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

An egg muffin is about 60 calories. A plate holds 4 egg muffins. About how many calories are on the plate?

(A) About 15
(B) About 64
(C) About 240
(D) About 600

HINT How many muffins, and how many calories in each one?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A large cereal box is 8 in long, 3 in wide, and 10 in tall. What is its volume?

(A) 21 cubic inches
(B) 80 cubic inches
(C) 240 cubic inches
(D) 240 square inches

HINT Volume of a box is length × width × height.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

Each jumbo box holds 400 cubic inches. How much do TWO jumbo boxes hold?

(A) 200 cubic inches
(B) 402 cubic inches
(C) 800 cubic inches
(D) 4,000 cubic inches

HINT Two equal boxes — what operation combines them?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which unit should the garden group use when reporting how much soil fills the beds?

(A) Feet
(B) Cubic feet
(C) Square feet
(D) Degrees

HINT Soil fills SPACE, not a line or a flat surface.

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

Yuzuki can buy three large boxes or two jumbo boxes for the same price. Give ONE reason in words (no equations) for either choice.

SENTENCE STARTER I recommend ___ because ___ .
A reason that is not about volume is ___ .

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

Why does it help to explain your thinking to others instead of just giving the answer?

SENTENCE STARTER Explaining helps because ___ .
When someone hears my reasons, they can ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

One of you sorted a card into 'Argument using equations' and the other into 'Argument using words.' Which argument would convince a skeptic faster, and what would it take to turn the other one into a counterexample?

SENTENCE STARTER An argument using ___ is stronger here because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 1.4

Math is Explaining and Sharing

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which statement is a CONJECTURE?

(A) "8 × 3 × 10 = 240."

(B) "I think the two jumbo boxes hold more cereal than the three large boxes."

(C) "Volume is measured in cubic units."

(D) "Yuzuki bought cereal."

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Three large boxes hold 240 cubic inches each. Two jumbo boxes hold 400 cubic inches each. Which is the correct volume comparison?

(A) 240 cubic inches vs. 400 cubic inches — the jumbo option holds more

(B) 480 cubic inches vs. 800 cubic inches — the jumbo option holds more

(C) 720 cubic inches vs. 800 cubic inches — the jumbo option holds more

(D) 720 cubic inches vs. 800 cubic inches — the large option holds more

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

One garden bed is 4 ft long, 3 ft wide, and 2 ft deep. A second bed is 6 ft long, 2 ft wide, and 2 ft deep. How much soil fills BOTH beds?

(A) 19 cubic feet

(B) 24 cubic feet

(C) 48 cubic feet

(D) 96 cubic feet

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The group found the volume by decomposing the garden into prisms and adding. Why would they compute it a SECOND, different way?

(A) Because the first way is always wrong

(B) To get a bigger number for the soil order

(C) If both ways give the same answer, the solution is accurate

(D) Because volume changes each time you measure it

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Answer in complete sentences.**

A student recommends the three large boxes because the cereal stays fresher — each box is open for less time. Is this argument convincing? Explain.

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

Use precise vocabulary to explain how you would find the soil needed for a bed 5 ft long, 2 ft wide, and 2 ft deep. Include the answer with its unit.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

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LESSON 1.4

Math is Explaining and Sharing

Challenge · Extension

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each prism to a DIFFERENT prism with the SAME volume.

_____ 8 × 3 × 10

_____ 4 × 3 × 2

_____ 5 × 8 × 10

_____ 6 × 2 × 4

(A) 6 × 4 × 10

(B) 6 × 2 × 2

(C) 20 × 10 × 2

(D) 8 × 3 × 2

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A student claims: “Greater box volume ALWAYS means more cereal inside.” Which is the best counterexample?

(A) A large box that costs more than a jumbo box

(B) A jumbo box with more volume that is sold half-filled with air

(C) Two jumbo boxes that are easier to store than three large boxes

(D) A box of a different cereal brand

SHOW YOUR WORK

3 Circle the letter of the best answer. Show how you know.

The garden needs 48 cubic feet of soil. One decomposition used two prisms of 24 cubic feet each. Which of these is ANOTHER decomposition of the same garden volume?

(A) A 3 × 3 × 3 prism and a 3 × 3 × 2 prism

(B) A 4 × 4 × 2 prism and a 4 × 2 × 2 prism

(C) A 4 × 4 × 2 prism and a 4 × 4 × 2 prism

(D) A 6 × 4 × 2 prism and a 2 × 2 × 2 prism

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 Answer in complete sentences.

Another garden bed design is also being considered. What information would you need to compare the two designs, and how would you argue for the one you recommend?

WRITE YOUR ANSWER

5 Answer in complete sentences.

Write TWO arguments for the same recommendation about Yuzuki's cereal — one using equations, one using only words. Then say which you think would convince a classmate more, and why.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

One of you sorted a card into 'Argument using equations' and the other into 'Argument using words.' Which argument would convince a skeptic faster, and what would it take to turn the other one into a counterexample?

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